

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
Fifth Semester B.Tech Degree Examination December 2021 (2019 scheme)

Course Code: HUT300

Course Name: INDUSTRIAL ECONOMICS & FOREIGN TRADE

Max. Marks: 100

Duration: 3 Hours

PART A

(Answer all questions; each question carries 3 marks)

1 What are the basic economic problems?

In every economy, some fundamental problems arise due to the scarcity of resources in relation to its demand. This gives rise to the problem of choice related to the allocation of resources to alternative uses. The problems are common and fundamental to all economies, whether they are rich or poor, developed or developing, socialist or capitalist. So they are called the central problems of an economy. These are:

1. The problem of allocation of resource: This problem arises due to the scarcity of resources in relation to its demand. It leads to the following questions;
 - a. What to produce? (It means What type of goods are to be produced?)
 - b. How to produce? (It means which technique is to be adopted ?)
 - c. For whom to produce? ? (It means the problem of distribution of goods)
2. The problem of fuller utilization of resource
3. The problem of growth of resource

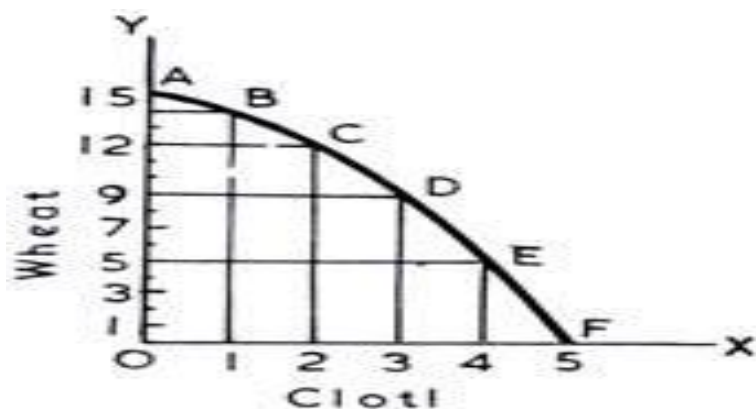
2 Explain Production Possibility Curve.

PPC or PPF is a graph that shows the different combinations of the quantities of two goods that can be produced or consumed in an economy, subject to limited availability of resources and a given technique of production. The PPC curve for an individual demonstrates the different combinations of two goods that the individual can have, with a given amount of income and at given price of the goods. Every choice made would have an opportunity cost, and the individual can get more units of one commodity only by giving up some units of the other commodity.

In the same sense, it can be for an individual firm and also for the economy as a whole.

Assumptions to illustrate the PPC for an economy:

1. Operating at full employment
2. Factors of production are fixed in supply
3. Technology remains the same



Production Possibility Curve

Fig. 21.2

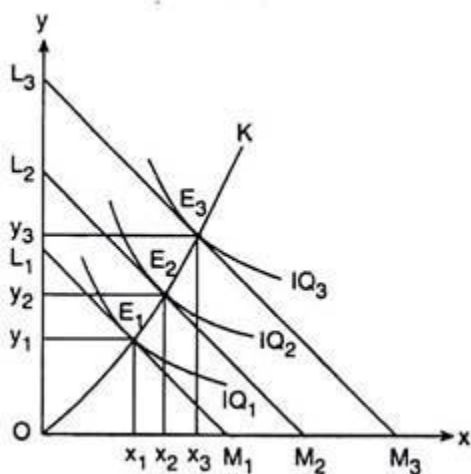
Properties of PPC

1. Slopes downwards from left to right: In a situation of fuller and efficient utilization of given resources, production of both the goods cannot be increased simultaneously. More of good x can be produced only with less of good y.
2. PPC is concave to origin: it implies that for every additional unit of good x produced, more and more of good y is to be sacrificed. It shows that the marginal opportunity cost or marginal rate of transformation or the slope of PPC tends to rise, as the resources are shifted from the production of good y to the production of good x.

3. Define expansion path.

Expansion path is a line connecting optimal input combinations as the scale of production expands. It gives the least cost input combinations at every level of output. We can obtain the expansion path by joining the point of tangency between isoquants and isocost lines of a firm.

- An expansion path is shown in the following diagram



4. What is margin of safety .What happens when margin of safety is low?

It is the sales beyond BEP. It is calculated as the difference between total sales and the Break even sales. It can be expressed in monetary terms or number of units. It can be expressed as below.

Margin of Safety = Sales – Break Even Sales

If margin of safety is large, which indicates that BEP is much below the actual sales, that means business is in sound condition and reduction in sales will not be a problem for the business. On the other hand, if margin of safety is low, any loss of sales may be a serious matter. Thus efforts need to be made to reduce fixed costs, variable costs or increasing the selling price or sales volume to improve contribution.

5. What is collusive oligopoly?

Oligopoly can either be collusive or non-collusive. Collusive oligopoly is a market situation wherein the firms cooperate with each other in determining price or output or both. A non-collusive oligopoly refers to a market situation where the firms compete with each other rather than cooperating.

Sometimes, firms may try to remove uncertainty related to acting independently and enter into price agreements with each other. This is collusion. Collusion is either formal or informal. It can take the form of cartel or price leadership. A cartel is an association of independent firms within the same industry which follow the common policies relating to price, output, sale, profit maximization, and the distribution of products. Price leadership is based on informed collusion. Under price leadership, one firm is a large or dominant firm and acts as the price leader who fixes the price for the products while the other firms allow it.

6. What is price skimming?

Price skimming is a product pricing strategy by which a firm charges the highest initial price that customers will pay and then lowers it over time. As the demand of the first customers is satisfied and competition enters the market, the firm lowers the price to attract another, more price-sensitive segment of the population.¹ The skimming strategy gets its name from "skimming" successive layers of cream, or customer segments, as prices are lowered over time.

Price skimming is often used when a new type of product enters the market. The goal is to gather as much revenue as possible while consumer demand is high and competition has not entered the market.

Once those goals are met, the original product creator can lower prices to attract more cost-conscious buyers while remaining competitive toward any lower-cost copycat items entering the market. This stage generally occurs when sales volume begins to decrease at the highest price the seller is able to charge, forcing them to lower the price to meet market demand. Generally, this technique is better-suited for lower-cost items, such as basic household supplies, where price may be a driving factor in most customers' production selections. Firms often use skimming to recover the cost of development. Skimming is a useful strategy in the following contexts:

- There are enough prospective customers willing to buy the product at a high price.
- The high price does not attract competitors.
- Lowering the price would have only a minor effect on increasing sales volume and reducing unit costs.
- The high price is interpreted as a sign of high quality.

7. How is GDP calculated?

Gross domestic product (GDP) is the standard measure of the value of final goods and services produced by a country for a particular time. Simply put, it is the market value of all the final goods and services produced within a nation in a given time. Here, market value means the use of market prices to value production. Final goods and services are produced for their final user and not for intermediate consumption

The calculation of **GDP** can be done on a quarterly, monthly, half-yearly, or annual basis. In India, GDP is computed at the Country level (Nations GDP), State level (Gross State Domestic Product), and the district level (District Gross Domestic Product). Commonly, GDP is measured in domestic currencies.

GDP is a measure of the economic health of a country. A negative GDP growth reflects bad signals for the economic status. Economists analyze GDP to figure out whether the economy is under recession, depression, or boom. It puts forward a sum of a country's production, in other words, the aggregate production level of an economy. It can also be used as an indicator for most governments and economic decision-makers for planning and devising various policies. GDP helps the investors to manage their portfolios by offering them guidance regarding the state of the economy. While framing fiscal and monetary policies, policymakers take note of GDP when contemplating decisions on interest rates, trade policies, and tax

8. Distinguish between Money market and Capital market.

Differences between Money Market and Capital Market

Money Market	Capital Market
Definition	
A random course of financial institutions, bill brokers, money dealers, banks, etc., wherein dealing on short-term financial tools are being settled is referred to as Money Market.	A kind of financial market where the company or government securities are generated and patronised with the intention of establishing long-term finance to coincide with the capital necessary is called Capital Market.
Market Nature	
Money markets are informal in nature.	Capital markets are formal in nature.
Instruments involved	
Commercial Papers, Treasury Certificate of Deposit, Bills, Trade Credit, etc.	Bonds, Debentures, Shares, Asset Secularisation, Retained Earnings, Euro Issues, etc.
Investor Types	

Commercial banks, non-financial institutions, central bank, chit funds, etc.	Stockbrokers, insurance companies, Commercial banks, underwriters, etc.
Market Liquidity	
Money markets are highly liquid.	Capital markets are comparatively less liquid.
Risk Involved	
Money markets have low risk.	Capital markets are riskier in comparison to money markets.
Maturity of Instruments	
Instruments mature within a year.	Instruments take longer time to attain maturity
Purpose served	
To achieve short term credit requirements of the trade.	To achieve long term credit requirements of the trade.
Functions served	
Increasing liquidity of funds in the economy	Stabilising economy by increase in savings
Return on investment achieved	
ROI is usually low in money market	ROI is comparatively high in capital market

9. What is free trade?

In the simplest of terms, free trade is the total absence of government policies restricting the import and export of goods and services. Free trade is the unrestricted importing and exporting of goods and services between countries.. The opposite of free trade is protectionism—a highly-restrictive trade policy intended to eliminate competition from other countries. In reality, however, governments with generally free-trade policies still impose some measures to control imports and exports.

Advantages of Free Trade

- **It stimulates economic growth:** Even when limited restrictions like tariffs are applied, all countries involved tend to realize greater economic growth.

- **It helps consumers:** Trade restrictions like tariffs and quotas are implemented to protect local businesses and industries. When trade restrictions are removed, consumers tend to see lower prices because more products imported from countries with lower labor costs become available at the local level.
- **It increases foreign investment:** When not faced with trade restrictions, foreign investors tend to pour money into local businesses helping them expand and compete. In addition, many developing and isolated countries benefit from an influx of money from U.S. investors.
- **It reduces government spending:** Governments often subsidize local industries, like agriculture, for their loss of income due to export quotas. Once the quotas are lifted, the government's tax revenues can be used for other purposes.
- **It encourages technology transfer:** In addition to human expertise, domestic businesses gain access to the latest technologies developed by their multinational partners.

5 Disadvantages of Free Trade

- **It causes job loss through outsourcing:** Tariffs tend to prevent job outsourcing by keeping product pricing at competitive levels. Free of tariffs, products imported from foreign countries with lower wages cost less. While this may be seemingly good for consumers, it makes it hard for local companies to compete, forcing them to reduce their workforce.
- **It encourages theft of intellectual property:** Many foreign governments, especially those in developing countries, often fail to take intellectual property rights seriously. Without the protection of patent laws, companies often have their innovations and new technologies stolen, forcing them to compete with lower-priced domestically-made fake products.
- **It allows for poor working conditions:** Similarly, governments in developing countries rarely have laws to regulate and ensure safe and fair working conditions. Because free trade is partially dependent on a lack of government restrictions, women and children are often forced to work in factories doing heavy labor under grueling working conditions.
- **It can harm the environment:** Emerging countries have few, if any environmental protection laws. Since many free trade opportunities involve the exporting of natural resources like lumber or iron ore, clear-cutting of forests and un-reclaimed strip mining often decimate local environments.
- **It reduces revenues:** Due to the high level of competition spurred by unrestricted free trade, the businesses involved ultimately suffer reduced revenues. Smaller businesses in smaller countries are the most vulnerable to this effect.

10. What is Devaluation ?

Devaluation is the deliberate downward adjustment of the value of a country's money relative to another currency, group of currencies, or currency standard. Countries that have a fixed exchange rate or semi-fixed exchange rate use this monetary policy tool. It is often confused with depreciation and is the opposite of revaluation, which refers to the readjustment of a currency's exchange rate.

The government issuing the currency may decide to devalue a currency. Devaluing a currency reduces the cost of a country's exports and can help shrink trade deficits. Unlike depreciation, it is not the result of nongovernmental activities. One reason a country may devalue its currency is to combat a trade imbalance. Devaluation reduces the cost of a country's exports, rendering them more competitive in the global market, which, in turn, increases the cost of imports. If imports are more expensive, domestic consumers are less likely to purchase them, further strengthening domestic businesses. Because exports increase and imports decrease, there is typically a better balance of

payments because the trade deficit shrinks. In short, a country that devalues its currency can reduce its deficit because there is greater demand for cheaper exports.

PART B

(Answer one full question from each module, each question carries 14 marks)

Module -1

11 a) List out the advantages and disadvantages of Joint Stock Company.

Most notable form nowadays

Formation: an entrepreneur conceives the idea of starting a business. He secures the assistance of some other persons. Then they decide the nature, size locality etc. and these initial persons behind the business are known as promoters of the company.

Registration: Incorporation of JSC under the Indian Companies Act 1956. Some legal formalities for it, first of all to prepare 2 documents

1. Memorandum of Association(MoA): includes name, registered office, object, liability and capital clause.
2. Articles of Association (AoA) : includes rules and regulations of the company.

These two documents should be submitted before the Registrar of companies, if he is satisfied, he issues a certificate of Incorporation.

Types: 2 types

1. Unlimited JSC: in which liability of the members not limited. So even their private property is liable.
2. Limited JSC: it may be of the following 2 types
 - a. Private Limited: minimum no. of shareholders is 2 and maximum is 50. its capital by members themselves, they cannot invite the general public to subscribe to its shares or debentures. They can start business immediately after the incorporation of the business, no need to issue a prospectus, no necessity to submit certain annual statements to the Registrar of companies.

Characteristics of Joint Stock Company

1. **An artificial person:** A joint stock company is an artificial person which is created by the law. It has no physical shape as a natural person but has almost all the rights of a natural person. It can sue others and can be sued.
2. **Perpetual existence:** A joint stock company is established by the law and the law brings it to an end. So, many shareholders may transfer their share and the new person may come in their place but it does not affect the existence of the company.
3. **Limited liability:** The liability of the shareholder is limited to the extent of the value of shares held or the amount guaranteed by them. If the company is unable to pay to the creditors then the shareholder won't pay anything more than what is to be paid to the company.
4. **Common seal:** A joint stock company is an artificial person. So, it cannot sign any contract in its name. Therefore, all the documents and contract papers require the affixing of the seal. Any documents with the common seal are only taken into consideration.
5. **Democratic management:** A joint stock company is a democratic organization. The important decisions are taken by following the principles of democracy in the annual general meeting and the board of director meeting.

6. **Transferability of shares:** The shares of the joint stock company can be transferable from one person to another without prior permission of the company management. They are free to transfer their shares.
7. **Capital is divided into shares:** A joint stock company divides its capital into a large number of parts with each value where each part of capital is called share. These shares are purchased by the general public as well as the promoters to be the shareholders of the company.
8. **Publication of financial statements:** A joint stock company should publish the audited financial statements yearly in a renowned national newspaper. This statement helps to provide information to the general public and other stakeholders.
9. **Separate legal entity:** A joint stock company is an artificial company so it has its own separate legal entity from its members. It can own assets, property, enter into contracts, sue or can be sued by anyone in the court by the law. Its shareholders cannot be held liable for any conduct of the company.

Merits of Joint Stock Company

1. **Huge capital:** A joint stock company has an association with various persons. It has the merits of huge capital because different member invests a large amount of capital. When there is a lack of capital in a joint stock company it can issue the shares to the public. Hence, huge capital can be collected when shares are issued.
2. **Perpetual existence:** A Joint stock company has a separate legal existence. The life of a joint stock company is not affected by death, lunacy, and insolvency of the members. Therefore, a joint stock company has a long-term life. Even if there are any changes in management, the board of directors or some member may come or go, the function of the company is not affected.
3. **Limited Liability:** Limited liability is the significance of a joint stock company. The shareholders should not pay the excess debt of the company by selling their private/personal property. Shareholders are liable up to the invested amount. Due to the provision of limited liability potential investors are attracted by a joint stock company.
4. **Transfer of shares:** A joint stock company has the provision of free transfer of shares. No one is compelled to join and leave the company. Permission or mutual consent is not needed to transfer the shares of a joint stock company.
5. **Democratic management:** A joint stock company has democratic management. It is managed by the majority of shareholders who elect the board of directors. They are responsible for managing the activities of a joint stock company. Competent members are elected from election to manage the company. Thus, democratic management can be seen in a joint stock company.
6. **Public faith:** People have faith in a joint stock company. It has an obligation to disclose the financial documents to the Annual General Meeting. The banks and the financial institutions believe in a joint stock company because of the accounts disclosed.
7. **Large-scale operation:** A Joint stock company has an association with different managerial skill because different members are associated with it. Due to the sufficient capital and competency of the members (directors) joint-stock company has the possibility of a large-scale operation. Hence, a joint stock company has a large-scale operation.
8. **Social importance:** A joint stock company is also a social creature. So, it invests amount for the betterment of society. It invests its capital in various sectors such as health, education, sports and so on. Hence, a joint stock company has a responsibility to society.

Demerits of Joint Stock Company

1. **Difficult legal formalities:** A joint stock company has demerits of difficult legal formalities. It is difficult to establish and run. It has to follow difficult legal formalities in comparison to sole

trading and partnership firm. It is rigidly observed by rules, regulations or laws of government because it collects capital from the general public.

2. **Lack of secrecy:** A Joint stock company has an obligation to disclose the accounts to insider as well as an outsider. The planned policies and strategies of the joint stock company are transparent because they are discussed in the Annual General Meeting. The company has to publish its statements of financial affairs every year. So, it has demerits of lack of secrecy.
 3. **Delay in decision making:** Sometimes a business organization has to take the quick decision but the quick decision is not possible in a joint stock company. The major decision of the company must be passed from the Annual General Meeting. So, long time is required to pass the decision. Thus, business delays grabbing the opportunities of an external environment.
 4. **Speculation of share:** There is the possibility of speculation of share in a joint stock company. Some shareholders have an inside approach with directors. Those shareholders can take undue advantage when they misuse the inside approach with directors.
 5. **Management of oligarchy:** Management oligarchy means the rule of the minority. The shareholders elect few directors in annual general meeting. Those few directors rule/control over the activities of a large number of shareholders. This is known as oligarchy.
 6. **Conflict of interest:** Different parties are involved in a joint stock company. They are shareholders, creditors/bankers, and employees. Different parties have different interest. If the interest of one party is addressed, it negatively impacts the interest of others. Therefore, a joint stock company has demerits of conflict of interest.
 7. **Neglect the minority:** A joint stock company gives priority to majority shareholders. Minority shareholders are neglected in Annual General Meeting and election. Only majority shareholders are given priority to take the decision of the joint stock company. Minority shareholders are boycotted to run productive business activities.
 8. **Groupism:** Unhealthy groups can be seen in the joint stock company in the period of election. Constructive groups are fruitful but destructive groups is harmful to the company. Unhealthy groups lead to the failure of the company. So, a joint stock company should be able to minimize unhealthy groups.
- b) **Define cross elasticity of demand. A tea manufacturing company was able to sell 8000 kg of tea when the price of coffee was Rs.70 per kg. Later they were able to sell 9000 kg when the price of coffee became Rs.80 per kg. Calculate the cross elasticity of demand for tea. Are the two commodities substitutes or complements? Give reason.**

Cross Elasticity Of Demand

It means rate of change in quantity demanded as a result of change in the price of a related commodity.

$$C E \text{ of } DD \text{ for } X \text{ and } Y = \frac{\% \text{ change in quantity demanded of } X}{\% \text{ change in price of } Y}$$

Assuming that the P of the commodity and income of the consumer as constant.

This type of E arises in the case of interrelated goods, such as substitutes and complimentary goods.

Two goods are complimentary, when the fall in price of Y increases the dd for X, eg. Petrol and car. And conversely, a rise in P of Y decreases the dd for X. Here E is negative.

Two goods are substitutes, when the fall in price of Y decreases the dd for X, eg. Tea and coffee. And conversely, a rise in P of Y increases the dd for X. Here E is positive.

$$(1000/8000) / (10/70) = 0.875:$$

Substitutes since cross elasticity is positive

OR

12 a) Define price elasticity of demand. A company producing soft drink is selling its product for Rs. 22. It sells 1000 units, and then increases the price to Rs.24. Now sales fall to 900 units. What is the price elasticity of soft drink? Is the demand elastic or inelastic? Why?

Price elasticity of demand: The attribute or quality of dd by virtue of which it increases or decreases under the pressure of a change in price is known as elasticity of dd. It expresses the degree of correlation between p and dd. It is the rate at which quantity demanded varies with a change in price. It refers to the sensitiveness or responsiveness of dd to a change in p.

The law of dd just states that a change in p leads to a change in dd, but does not explain the rate of change. This rate of change may be slow or rapid . E of dd is different for different goods and services and it may not be the same for the same commodity at all times.

- There are five degrees of e of dd.
- 1. Perfectly elastic or infinite e of dd.
- 2. Elastic or more elastic dd.
- 3. Unitary elastic dd.
- 4. Less elastic or inelastic dd
- 5. Perfectly inelastic or zero elasticity of dd

$$(100/1000) / (2/22) = -1.1$$

Demand is elastic since the value of price elasticity is greater than 1

b) With the help of diagram explain Deadweight loss

Deadweight loss can be defined as an economic inefficiency that occurs as a result of a policy or an occurrence within a market, that distorts the equilibrium set by the free market. These inefficiencies affect both the demand and supply sides of the market in question. An economic inefficiency refers to a situation where the quantity of goods or services being supplied and the price that it is being exchanged for is not equal to the equilibrium set by the supply and demand components, of that market.

A deadweight loss, in economics, can be caused by multiple policies and inefficiencies within a market. Some of those causes are listed below:

- **Price ceilings**
This is when a government mandates that the price of a certain good or service should not exceed a particular value. This is most commonly seen in the renting market, where an upper limit is placed on what a landlord may charge for rent. This results in demand outstripping supply and a deadweight loss manifesting.
- **Price floors**
This is when a government instates a minimum price that may be charged for a specific good or service. The most common example of this is the minimum wage.
- **Taxation**
Sales taxes refer to a specific category of taxes and are levied upon the value of goods and services. This results in prices for goods and services being elevated without an increase in demand or a decrease in supply. This results in both supply and demand decreasing, since consumers are paying

more and producers are receiving a lower percentage of their revenue, and a deadweight loss occurs.

- **Monopoly**

A monopoly refers to a situation where one firm is the sole provider of a good or service, to a particular market. This results in the monopolist charging a higher price than would be the case in a competitive market. This also results in the number of goods or services sold being less than what is optimal (where marginal revenue is equal to marginal cost). This results in a deadweight loss for society.

- **Subsidies**

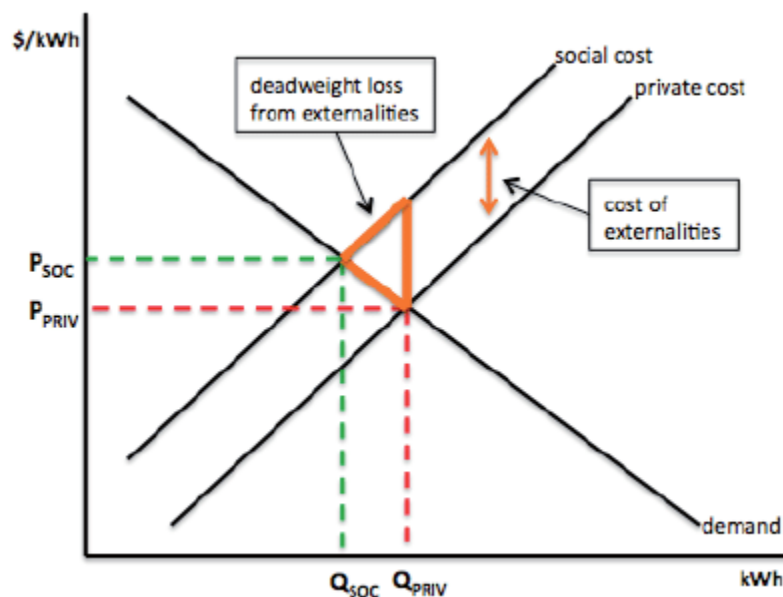
Subsidies are the opposite of taxes and refer to government initiatives to make goods and/or services more affordable to the public. This will result in the consumer having to pay less, a producer receiving more for their goods/services, and the government making up the difference. This results in consumer and producer surpluses increasing, but the added loss to the government will result in the aggregate effect, on society, being a deadweight loss.

- **Product surplus**

If there are more goods in an economy than demand for it, it can result in a deadweight loss. This is because that extra money and resources are being tied up in these dormant products, rather than flowing through the economy.

- **Product deficit**

If there are not enough goods to satisfy demand, then this results in an opportunity loss. This is because fewer products are being sold than would in an efficient market and thus a deadweight loss occurs.



Module -2

13 a) Explain Law of variable proportion with a diagram.

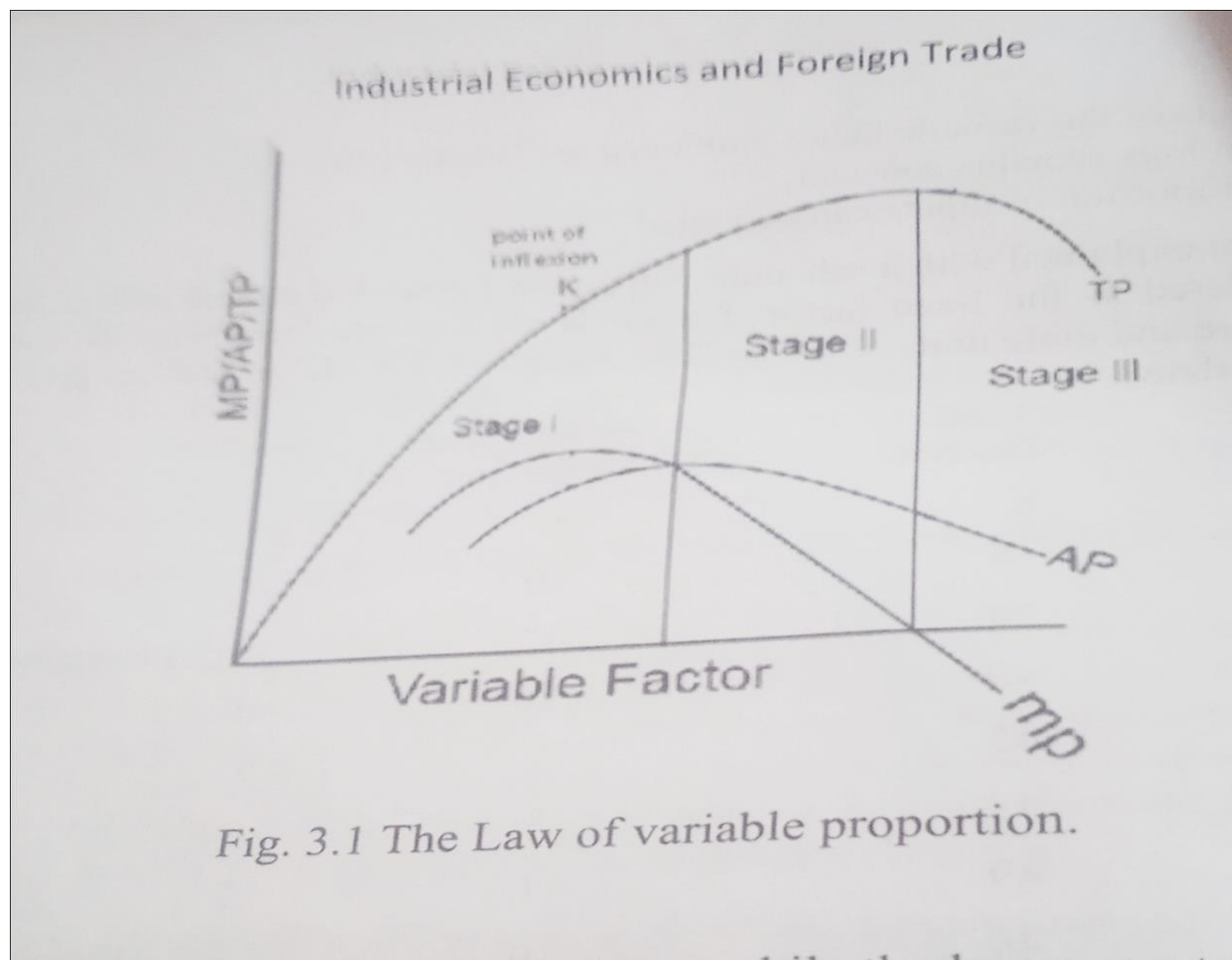
The law shows the production function with one input factor variable while keeping the other input factors constant. It states that if one factor is used more and more (variable), keeping the other factors constant, the total output will increase at an increasing rate in the beginning and then at a diminishing rate and eventually decreases provided there is no change in technology.

Assumptions of the law

1. The state of technology does not change.
2. Only one factor is made variable and other factors are kept constant.
3. The variable factor units are homogenous.
4. A particular product can be produced under varying proportions of the input combinations
5. The law operates in the short run.

Units of fixed factors(land, tools, seeds, manures etc.)	Units of labour	TP	MP	AP
1	1	2	-	2
1	2	5	3	2.5
1	3	9	4	3
1	4	12	3	3
1	5	14	2	2.8
1	6	15	1	2.5
1	7	15	0	2.1
1	8	14	-1	1.7

- TP first rises at an increasing rate upto 3 units of labour, afterwards it rises at decreasing rate. When 6 labourers are employed, the TP is maximum. It begins to decline when 8 labourers are employed.
- MP first increases reaching the maximum point when 3 labourers are employed. Afterwards it starts declining but remains positive upto the employment of 6th labourer. The MP of the 7th unit of labourer is zero and 8th unit is negative.
- AP of labour first increases and then declines gradually. AP becomes equal to MP when the latter starts declining. The MP drops off faster than the AP.



Three stages of production

As per the law, the behaviour of output when varying the quantity of one factor combined with fixed quantities of the others can be divided into 3 distinct stages:

Stage I -Stage of increasing returns: all the 3, i.e. TP, MP and AP are increasing. The reason is that the efficiency of fixed factors increases as additional units of the variable factors are added to it. MP in this stage increases but in a later part it starts declining. Though MP starts declining, it is greater than the AP and the AP continues to rise. Stage 1 ends when the AP reaches its maximum.

Stage II -Stage of decreasing returns: the TP continues to increase but at a diminishing rate. The reason is that the fixed factors become inadequate relative to the quantity of the variable factor. MP and AP are declining, but are positive. At the end of the second stage, TP is maximum and the MP is zero. Stage II ends at the point where the MP is zero.

Stage III -Stage of negative returns: TP starts to decline. This is due to variable factor is too much relative to the fixed factor. AP shows a steady decline, but never becomes zero. MP becomes negative and so known as the stage of negative returns.

Reasons for the operation of the law

1. Indivisibility of factors

2. Change in factor ratio
3. imperfect substitutes
4. Excessive variable factor

b) Explain Marginal revenue and Average revenue in Perfect competition and imperfect competition with graph.

Perfect competition is a form of the market in which there is a large number of buyers and sellers and where homogeneous product is sold at a uniform price. A price taker firm means that it has to accept the price as determined by the forces of market demand and market supply. Firm's demand curve under perfect competition is a horizontal straight line parallel to X-axis. Under perfect competition, AR is constant for a firm. Hence, $AR = MR$.

A firm under perfect competition is price-taker. This simply means it can alter its volume of output and sales level without significantly affecting the market price of its product. This explains why a firm operating in a perfectly competitive market has no power to influence that market through its own individual actions. It must passively accept whatever price happens to prevail in the market. At the prevailing market (ruling) price it can sell as much as it likes. This means that the demand for its product is completely elastic at a particular (market determined) price.

The relationship between AR and MR of a firm under **imperfect competition** is given below: Both Monopoly and Monopolistic Competition fall under the category of Imperfect Competition. Therefore, AR and MR curves slope downwards as more units can be sold only by reducing the price. However, there is one major difference between AR and MR curves of monopoly and monopolistic competition. Under monopolistic competition, the AR and MR curves are more elastic as compared to those of Monopoly. It happens because of the presence of close substitutes under monopolistic competition and the absence of close substitutes under monopoly. So, when the price of a commodity is increased in both the markets, then proportionate fall in demand under monopoly is less than proportionate fall in demand under monopolistic competition. Read more on Sarthaks.com - <https://www.sarthaks.com/823346/with-the-help-diagram-explain-the-relationship-between-firm-under-imperfect-competition>

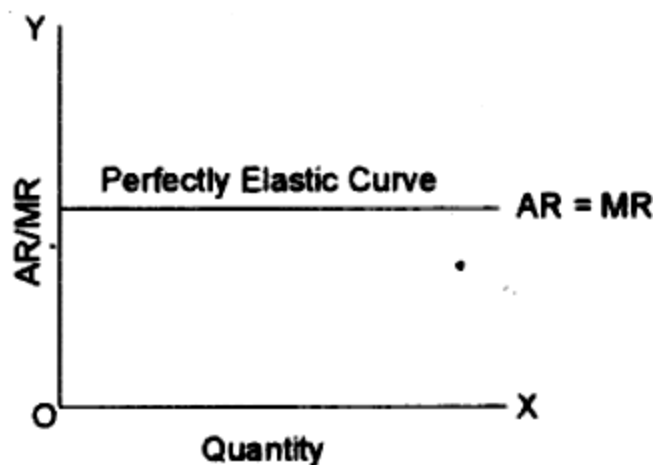


Fig. Under Perfect Competition

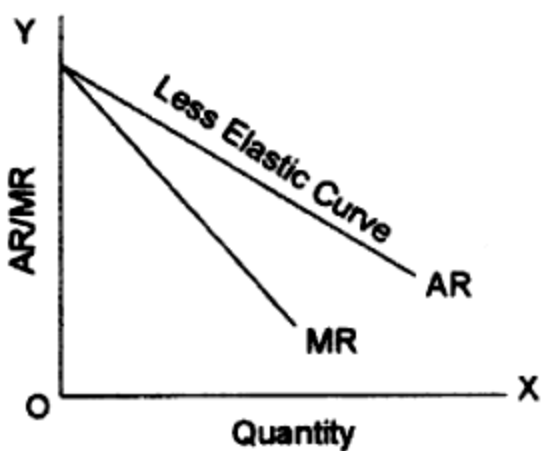


Fig. Under Imperfect Competition

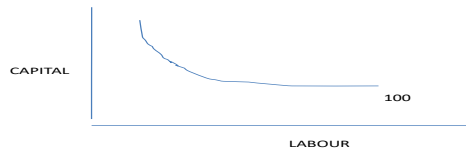
OR

14 a) Define Isoquant curve. Explain the properties of Isoquant curve.

An iso quant is a curve which shows various combinations of two inputs which give the same level of output. Iso means equal and quant means quantity. Also called iso product curves or equal product curves. It can be explained with the help of the following table and diagram

Combination of labour and capital	Units of labour (L)	Units of capital (K)	Output of cloth in meters
A	5	9	100
B	10	6	100

C	15	4	100
D	20	3	100



Properties of an Iso Quant

1. Iso quants are negatively sloped
2. Isoquants are convex o the origin
3. Two isoquants never intersect
4. Higher iso quants represents higher levels of output

- b) **The total sales of a manufacturing firm is Rs 20000 in this year. Its variable costs are Rs 8000 while its fixed costs are Rs 6000 for that year. Find the breakeven point of this firm.**

$$P/V \text{ ratio} = C/S = (S - V)/S = 12000/20000 = 0.6$$

$$\text{Breakeven point} = F/PV \text{ ratio} = 6000/0.6 = \text{Rs } 10000$$

Module -3

15 a) Distinguish between monopoly and oligopoly.

A monopoly and an oligopoly are market structures that exist when there is imperfect competition. A monopoly is when a single company produces goods with no close substitute, while an oligopoly is when a small number of relatively large companies produce similar, but slightly different goods. In both cases, significant barriers to entry prevent other enterprises from competing.

Monopoly

A monopoly exists in areas where one company is the only or dominant force to sell a product or service in an industry. This gives the company enough power to keep competitors away from the marketplace. This could be due to high barriers to entry such as technology, steep capital requirements, government regulation, patents or high distribution costs.

Once a monopoly is established, lack of competition can lead the seller to charge high prices. Monopolies are price makers. This means they determine the cost at which their products are sold. These prices can be changed at any time. A monopoly also reduces available choices for buyers. The monopoly becomes a pure monopoly when there is absolutely no other substitute available.

Monopolies are allowed to exist when they benefit the consumer. In some cases, governments may step in and create the monopoly to provide specific services such as a railway, public transport or postal services. For example, the United States Postal Service enjoys a monopoly on first class mail and advertising mail, along with monopoly access to mailboxes

Oligopoly

In an oligopoly, a group of companies (usually two or more) controls the market. However, no single company can keep the others from wielding significant influence over the industry, and they each may sell products that are slightly different.

Prices in this market are moderate because of the presence of competition. When one company sets a price, others will respond in fashion to remain competitive. For example, if one company cuts prices, other players typically follow suit. Prices are usually higher in an oligopoly than they would be in perfect competition.

Because there is no dominant force in the industry, companies may be tempted to collude with one another rather than compete, which keeps non-established players from entering the market. This cooperation makes them operate as though they were a single company.

Examples of Monopolies and Oligopolies

A company with a new or innovative product or service enjoys a monopoly until competitors emerge. Sometimes these new products are protected by law. For example, pharmaceutical companies in the U.S. are granted 20 years of exclusivity on new drugs.⁷ This is necessary due to the time and capital required to develop and bring new drugs to market. Without this protected status, firms would not be able to realize a return on their investment, and potentially beneficial research would be stifled.

Gas and electric utilities are also granted monopolies. However, these utilities are heavily regulated by state public utility commissions. Rates are often controlled, along with any rate increases the company may pass onto consumers.

Oligopolies exist throughout the business world. A handful of companies control the market for mass media and entertainment. Some of the big names include The Walt Disney Company (DIS), ViacomCBS (VIAC) and Comcast (CMCSA). In the music business, Universal Music Group and Warner Music Group have a tight grip on the market.

b) With the help of a diagram, explain equilibrium under monopolistic competition.

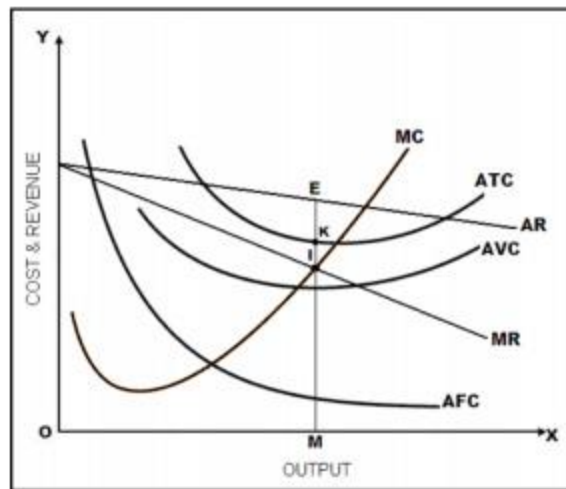
A Firm's Short-Run Equilibrium under Monopolistic Competition

Under Monopolistic Competition, the revenue curves are downward sloping (like under Monopoly). This is because, in order to sell more, the firm has to decrease the price.

A firm under Monopolistic Competition can either earn normal profits, super-normal profits, or incur losses. Also, like under Monopoly, a firm earns super-normal profits if the demand for its product is very high.

Also, in the short-run, new firms cannot enter the group and enhance the supply of the product group. Therefore, they cannot compete away the super-normal profits of the firm. Also, in the short-run, a firm faces certain fixed costs. These can include production as well as selling costs.

Short Run Equilibrium under Monopolistic Competition



In the figure above, you can see that the AR and MR curves of the firm have negative slopes. Further, the AVC curve includes the production costs as well as the variable components of selling expenses. Furthermore,

The MC curve cuts the AVC curve at its lowest point. Also, the ATC curve represents the average of the total cost of the firm including the fixed selling expenses.

The MC curve intersects the MR curve from below at point I. Hence, the firm decides to produce a quantity of OM and charge a price of EM per unit.

By doing so, the firm earns a profit of EK per unit and the entry of rival firms do not compete it out. However, based on the relative location of the cost and revenue curves, it is possible that the firm is in equilibrium with:

- Only normal profit
- Covering a part of fixed costs. Therefore, incurring a loss less than its fixed costs
- Loss equal to the fixed costs (where AR is tangent to the AVC curve)

A Firm's Long-Run Equilibrium under Monopolistic Competition

To discuss a firm's long-run equilibrium under Monopolistic Competition, it is important to remember the following points:

- There are no fixed costs in the long-run. The firm can vary its inputs as well as its selling costs. Further, the firm can choose between various product qualities.
- There is no compulsion on a firm to operate at a loss. It can leave the industry whenever it wants. When a firm leaves the industry, the absolute market shares of the remaining firms, increase. Further, their demand curve shifts right and upwards. This continues until other firms can produce without incurring a loss.
- On the other hand, if the demand is so strong that the existing firms make super-normal profits, then new firms can enter the group.
- They produce close substitutes of the existing products and increase the total product supply. Therefore, the demand shares of the existing firms reduce. Hence, the demand curve of a firm cannot stay above its long-run average cost curve.
- All firms operating under Monopolistic Competition can make a choice between combinations of:
 - Product quality
 - Product Differentiation

- Selling costs
- A firm must consider the fact that any variation of price on its part can attract a reaction from its rivals. Therefore, it faces a much steeper demand curve.

Therefore, under Monopolistic Competition, a firm is exposed to constant interaction with the rest of the firms in the group. Its decisions are not independent of the decisions of the other firms.

Further, the firm's demand curve depends on its actions AS WELL AS on the actions of its rivals. Therefore, it must consider different combinations of its cost components pertaining to the product quality and its selling expenses, etc. This helps the firm estimate the slope and position of the demand curve.

Graphical Representation

Fig.i: Long Run Equilibrium of Firm under Monopolistic Competition

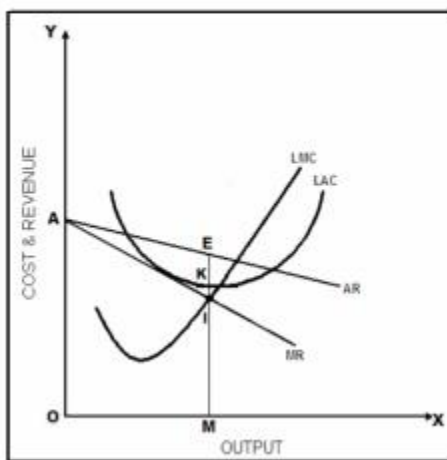
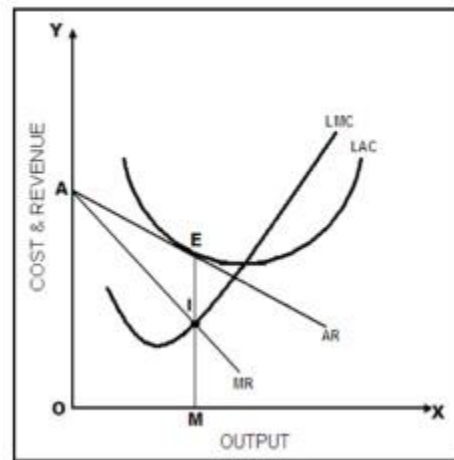


Fig. ii: Long Run Equilibrium of Firm under Monopolistic Competition



Let's say that the LAC curve in Fig. I represent the product quality and selling expenses that a firm selects. This has a corresponding long-term MC curve (LMC) which intersects the MR curve from below at point I.

Therefore, the firm decides to produce a quantity of OM and sell it a per unit price of EM. This gets a profit of EK per unit. However, soon new firms enter the market and start offering close substitutes and bring the profit down.

Therefore, there is a reduction in the market shares of the existing firms. The firm's AR curve shifts left until it becomes tangent to the LAC curve at point E as shown in Fig. ii. This ensures that the firm earns only normal profit. Once this stage is reached, there is no incentive for new firms to enter the market.

This results in the firm's long-term equilibrium under Monopolistic Competition. The equilibrium is given by the point of tangency between the firm's AR curve and LAC curve, which is at point E in Fig. ii. Therefore, in the long-run, under monopolistic competition, firms earn only normal profits.

16 a) Explain kinked demand curve model.

In many oligopolist markets, it has been observed that prices tend to remain inflexible for a very long time. Even in the face of declining costs, they tend to change infrequently. American economist Sweezy came up with the kinked demand curve hypothesis to explain the reason behind this price rigidity under oligopoly.

According to the kinked demand curve hypothesis, the demand curve facing an oligopolist has a kink at the level of the prevailing price. This kink exists because of two reasons:

1. The segment above the prevailing price level is highly elastic.
2. The segment below the prevailing price level is inelastic.

The following figure shows a kinked demand curve dD with a kink at point P.

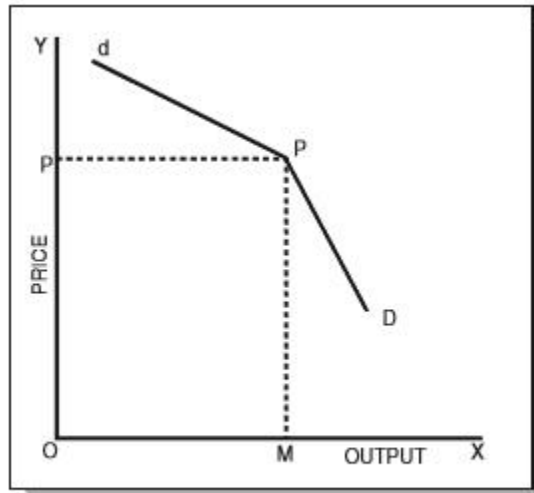


Fig. 1 : Kinked Demand Curve under oligopoly

From the figure, we know that

- i. The prevailing price level = P
- ii. The firm produces and sells output = OM
- iii. Also, the upper segment (dP) of the demand curve (dD) is elastic.
- iv. The lower segment (PD) of the demand curve (dD) is relatively inelastic.

This difference in elasticities is due to an assumption of the kinked demand curve hypothesis.

Assumption:

Each firm in an oligopoly believes the following two things:

- a. If a firm lowers the price below the prevailing level, then the competitors will follow him.
- b. If a firm increases the price above the prevailing level, then the competitors will not follow him.

There is logical reasoning behind this assumption. When an oligopolist lowers the price of his product, the competitors feel that if they don't follow the price cut, then their customers will leave them and buy from the firm who is offering a lower price. Therefore, they lower their prices too in order to maintain their customers. Hence, the lower portion of the curve is inelastic. It implies that if an oligopolist lowers the price, he can obtain very little sales. On the other hand, when a firm increases the price of its product, it experiences a substantial reduction in sales. The reason is simple – consumers will buy the same/similar product from its competitors. This increases the competitors' sales and they will have no motivation to match the price rise. Therefore, the firm that raises the price suffers a loss and hence refrain from increasing the price. This behavior of oligopolists can help us understand the elasticity of the upper portion of the demand curve (dP). The figure shows that if a firm raises the price of a product, then it experiences a large fall in sales. Hence, no firm in an oligopolistic market will try to increase the price and a kink is formed at the prevailing price. This is how the kinked demand curve hypothesis explains the rigid or sticky prices.

b) Which are the different types of Non price competition under Oligopoly

Non-price competition refers to competition between companies that focuses on benefits, extra services, good workmanship, product quality – plus all other features and measures that do not involve altering prices. It contrasts with price competition, in which rivals try to gain market share by reducing their prices.

Non-price competition is a marketing strategy that typically includes promotional expenditures such as sales staff, sales promotions, special orders, free gifts, coupons, and advertising.

Put simply, it means marketing a firm's brand and quality of products, rather than lowering prices. Most companies across the world are involved in either non-price competition, price competition, or both.

Non-price competition – oligopolies

Non-price competition is more common in markets where there is imperfect competition, such as those with very few competitors – oligopolies – maybe because it can give an impression of a very competitive market, when in fact the rivals are colluding to keep their prices high.

Non-price competition is an important strategy in marketplaces where sellers are offering their service as a product, such as AirBnB, Fiverr, oDesk, TaskRabbit, Mechanical Turk, etc.

Non-price competition – two phases

There are typically two phases to a non-price competition strategy. The first implements new aspects of production or services, while the second lets consumers know about them.

Non-price competition – pros & cons

There are several **advantages** associated with this type of marketing campaign:

- Better sales tactics, including social media posts, efficient forms of online advertising, and direct sales through the manufacturer.
- Improved product quality.
- Different presentation of products for varied demographics. For example, sales may increase if the same product is presented differently for men and women.
- Superior brand perception. Brands provide guidance and clarity for choices made by firms, consumers, investors and other stakeholders.

Disadvantage:

The main drawback is that consumers are not likely to notice the changes straight away – which is not the case when prices are lowered.

17 a) Explain Circular flow of income in two sector model and four sector model.

Circular Income Flow in a Two Sector Economy:

Real flows of resources, goods and services have been shown in the following figure. In the upper loop of this figure, the resources such as land, capital and entrepreneurial ability flow from households to business firms as indicated by the arrow mark.

In opposite direction to this, money flows from business firms to the households as factor payments such as wages, rent, interest and profits.

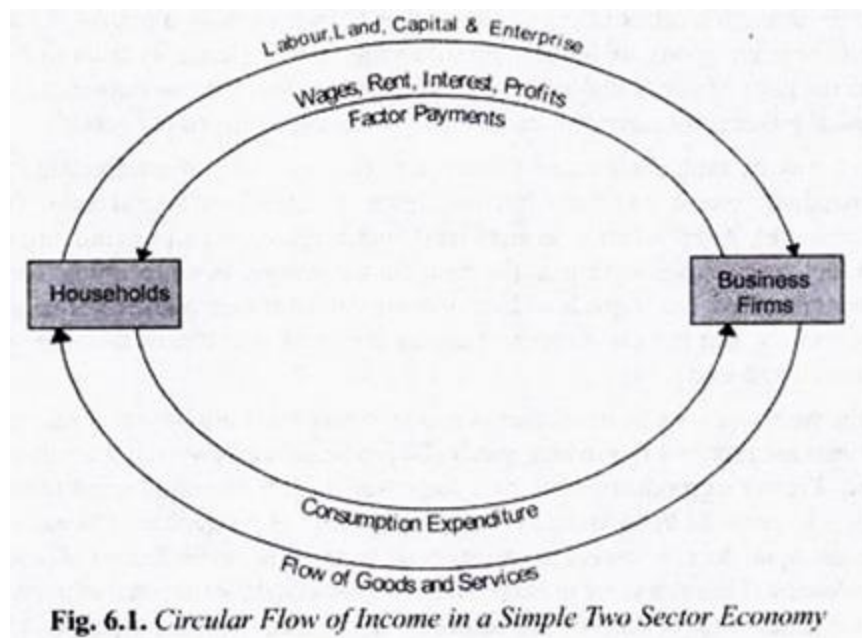


Fig. 6.1. Circular Flow of Income in a Simple Two Sector Economy

In the lower part of the figure, money flows from households to firms as consumption expenditure made by the households on the goods and services produced by the firms, while the flow of goods and services is in opposite direction from business firms to households.

Thus we see that money flows from business firms to households as factor payments and then it flows from households to firms. Thus there is, in fact, a circular flow of money or income. This circular flow of money will continue indefinitely week by week and year by year. This is how the economy functions. It may, however, be pointed out that this flow of money income will not always remain the same in volume.

In other words, the flow of money income will not always continue at a constant level. In year of depression, the circular flow of money income will contract, i.e., will become lesser in volume, and in years of prosperity it will expand, i.e., will become greater in volume.

This is so because the flow of money is a measure of national income and will, therefore, change with changes in the national income. In year of depression, when national income is low, the volume of the flow of money will be small and in years of prosperity when the level of national income is quite high, the flow of money will be large.

In order to make our analysis simple and to explain the central issues involved, we take many assumptions. In the first place, we assume that neither the households save from their incomes, nor the firms save from their profits. We further assume that the government does not play any part in the national economy.

In other words, the government does not receive any money from the people by way of taxes, nor does the government spend any money on the goods and services produced by the firms or on the resources and services supplied by the households. Thirdly, we assume that the economy neither imports goods and services, nor exports anything. In other words, in our above analysis we have not taken into account the role of foreign trade. In fact we have explained above the flow of money that occurs in the functioning of a closed economy with no savings and no role of government.

Circular Income Flow with Saving and Investment:

In our above analysis of the circular flow of income we have assumed that all income which the households receive, they spend it on consumer goods and services. A result, circular flow of money

speeding and income remains undiminished. We will now explain if households save a part of their income, how their savings will affect money flows in the economy.

When households save, their expenditure on goods and services will decline to that extent and as a result money flow to the business firms will contract. With reduced money receipts, firms will hire fewer workers (or lay off some workers) or reduce the factor payments they make to the suppliers of factors such as workers.

This will lead to the fall in total incomes of the households. Thus, savings reduce the flow of money expenditure to the business firms and will cause a fall in economy's total income. Economists therefore call savings a leakage from the money expenditure flow.

Circular Income Flow in a Three Sector Economy with Government:

In our above analysis of money flow, we have ignored the existence of government for the sake of making our circular flow model simple. This is quite unrealistic because government absorbs a good part of the incomes earned by households. Government affects the economy in a number of ways.

Here we will concentrate on its taxing, spending and borrowing roles. Government purchases goods and services just as households and firms do. Government expenditure takes many forms including spending on capital goods and infrastructure (highways, power, communication), on defence goods, and on education and public health and so on. These add to the money flows which are shown in Fig. 6.3 where a box representing Government has been drawn. It will be seen that government purchases of goods and services from firms and households are shown as flow of money spending on goods and services.

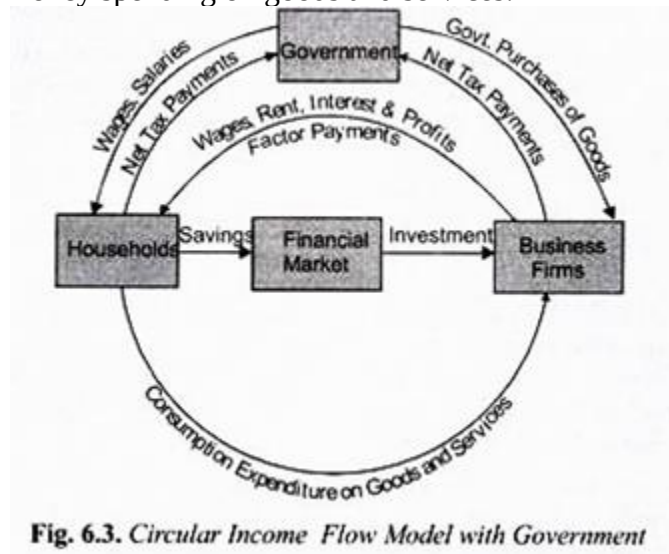


Fig. 6.3. *Circular Income Flow Model with Government*

Government expenditure may be financed through taxes, out of assets or by borrowing. The money flow from households and business firms to the government is labelled as tax payments in Fig. 6.3. This money flow includes all the tax payments made by households less transfer payments received from the Government. Transfer payments are treated as negative tax payments.

Money Income Flows in the Four Sector Open Economy: Adding Foreign Sector:

We now turn to explain the money flows that are generated in an open economy, that is, economy which have trade relations with foreign countries. Thus, the inclusion of the foreign sector will reveal to us the interaction of the domestic economy with foreign countries. Foreigners interact with the domestic firms and households through exports and imports of goods and services as well

as through borrowing and lending operations through financial market. Goods and services produced within the domestic territory which are sold to the foreigners are called exports. On the other hand, purchases of foreign-made goods and services by domestic households are called imports. Figure 6.4 illustrates additional money flows that occur in the open economy when exports and imports also exist in the economy. In our analysis, we assume it is only the business firms of the domestic economy that interact with foreign countries and therefore export and import goods and services.

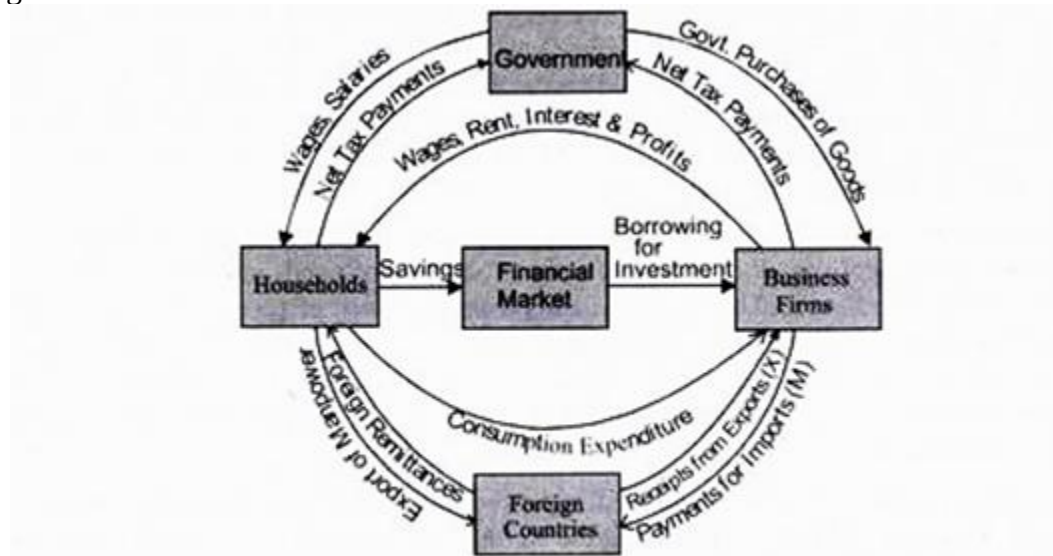


Fig. 6.4. Circular Flow of Income in an Open Economy with Government and Foreign Sector

A flow of money spending on imports have been shown to be occurring from the domestic business firms to the foreign countries (i.e., rest of the world). On the contrary, flow of money expenditure on exports of a domestic economy has been shown to be taking place from foreign countries to the business firms of the domestic economy.

If exports are equal to the imports, then there exists a balance of trade. Generally, exports and imports are not equal to each other. If value of exports exceeds the value of imports, trade surplus occurs. On the other hand if value of imports exceeds value of exports of a country, trade deficit occurs.

In the open economy there is interaction between countries not only through exports and imports of goods and services but also through borrowing and lending funds or what is also called financial market. These days financial markets around the world have become well integrated.

When there is a trade surplus in the economy, that is, when exports (X) exceed imports (M), net capital inflow will take place. By net capital inflow we mean foreigners will borrow from domestic savers to finance their purchases of domestic exports. In this way as a result of net capital inflow domestic savers will lend to foreigners, that is, acquire foreign financial assets.

On the contrary, in case of import surplus, that is, when imports are greater than exports, trade deficit will occur. Therefore, in case of trade deficit, domestic consumer households and business firms will borrow from abroad to finance their excess of imports over exports. As a result, foreigners will acquire domestic financial assets.

From the circular flows that occur in the open economy the national income must be measured by aggregate expenditure that includes net exports, that is, $X - M$ where X represents exports and M

represents imports. Imports must be subtracted from the total expenditure on foreign produced goods and services to get the value of net exports. Thus, in the open economy

$$\text{National Income} = C + I + G + NX$$

where NX represents net exports, $X - M$.

Since national income can be either consumed, saved or paid as taxes to the Government we have $C + I + G + NX = C + S + T$

b) From the data given below estimate the NDP, using income method and expenditure method.

Items	(Rs .Crores)
Consumption expenditure	3000
Investment expenditure	2000
Government expenditure	700
Exports	600
Import	300
Intermediate Consumption	2000
Wages and salaries	2000
Rent	500
Interest	500
Profit	1000

Income method: $W + R + I + P = 2000 + 500 + 500 + 1000 = 4000$:

Expenditure method: $C + I + G + X - M = 3000 + 2000 + 700 + 600 - 300 = 6000 - 2000$ (Intermediate Cons.) = 4000

OR

18 a) What are the different methods used to control Inflation

Inflation

Meaning: increasing price level in the economy.

Definition

Crowther: “inflation is a state in which the value of money is falling, i.e. the prices are rising”.

R. G. Hawtrey: “inflation is the issue of too much currency”.

Prof. Coulbourn: “inflation is too much money chasing too few goods”.

H. G. Johnson: “inflation is a substantial rise in prices”.

Thus inflation means substantial and rapid rise in the general level of prices.

Features:

- Excessive rise in prices
 - Excessive money ss
 - Inflation breeds further inflation
- small levels of inflation are not bad, but when it leads to rise in wages and cost of production, and thereby further increase in prices, it should be kept under control.

Causes of inflation

1. Imbalance between total ss and total dd for goods
2. During war, as the govt, makes additional issue of notes
3. Govt. resorts to deficit financing

4. Black and unaccounted money and counterfeit money
5. Hoarding by traders and consumers
6. When wages and other costs of production rises at a rate higher than the increase in labour productivity (cost push inflation)
7. When increase in price comes from increased incomes and higher effective demand (demand pull inflation)

Control of inflation

1. Monetary measures:
 - a. Volume of money may be reduced.
 - b. Increased bank rate
 - c. Sale of govt. securities in the open market
 - d. Higher reserve requirements
 - e. Consumer credit control
 - f. Higher margin requirements
 - g. Over valuation of currency
2. Fiscal measures
 - a. Reduction in govt. expenditure
 - b. Increase in taxes
 - c. Public borrowing
 - d. Compulsory deposits
3. Others
 - a. Expansion of output
 - b. Wage policy towards wage freezing
 - c. Price control by way of rationing
 - d. Control over hoarding and black marketing
 - e. Import of essential goods

b) From the data given below estimate Gross National Product, Net National Product and National income.

GDP 5000 (in 100 billion)

NFIA -50

Indirect tax 70

Subsidies 20

Depreciation 30

Gross national product (GNP) refers to the total value of all the goods and services produced by the residents and businesses of a country, irrespective of the location of production. GNP takes into account the investments made by the businesses and residents of the country, living both inside and outside the country. It also takes into account the value of the products produced by the industries of the domestic origin. GNP does not take into consideration the incomes earned by the foreign nationals in the country or any products produced by a foreign company in the manufacturing units in the country.

Net national product or NNP is the market value of all the finished goods and services that are produced by citizens of a nation, living domestically and internationally during a year. Net national product is also referred to as the value that is obtained by subtracting depreciation from the gross national product (GNP). Net national product considers all the goods, products and services that are manufactured by the country's citizens, irrespective of their location, or in other words, net

national product considers products that are produced domestically and also from overseas. NNP is one of the important metrics for determining the actual growth of a nation. It measures how much the country is able to consume in a given period of time.

National income means the value of goods and services produced by a country during a financial year. Thus, it is the net result of all economic activities of any country during a period of one year and is valued in terms of money. National income is an uncertain term and is often used interchangeably with the national dividend, national output, and national expenditure.

GNP = 4950, NNP = 4920, NI = 4870

Module -5

19 a) What is international trade? List out the advantages of Foreign trade?

International trade is referred to as the exchange or trade of goods and services between different nations. This kind of trade contributes and increases the world economy. The most commonly traded commodities are television sets, clothes, machinery, capital goods, food, raw material, etc. International trade has exceptionally increased, which includes services such as foreign transportation, travel and tourism, banking, warehousing, communication, distribution, and advertising. Other equally important developments are the increase in foreign investments and production of foreign goods and services in an international country. foreign investments and productions help companies to come closer to their international customers, thus serving them with goods and services at a very low rate. All the mentioned activities are parts of international business. It can be concluded by saying that international trade and production are two aspects of international business, which is growing day by day across the globe.

Reasons for International Trade

(1) Production

It is not possible for every single country to produce equally at a cheap cost. That is why international trade is taken into account.

(2) Factors of production

Factors of production include labour, capital, and raw material for producing goods and services that are available at different rates in different countries.

(3) Cost of production

Each country finds it advantageous to produce only those goods and services that can be produced efficiently. The rest of the activities are assigned to other countries at a lower cost.

(4) Resource distribution

Many times, companies face problems due to the limitation of natural resources. There is an unequal distribution of the resources in the country.

Examples

Different countries are specialised in different sectors like in India, Maharashtra is involved in textiles, West Bengal in jute products, Haryana and Punjab in food products, Kerala in spices, etc.

Advantages of International Trade

International trade between various nations is an essential factor that is responsible for the increase in the standard of living, creating employment, and empowering consumers to enjoy different kinds of goods. Few other important factors that are influenced by international trade are:

Utilisation of raw materials: Some countries are naturally blessed with an abundance of raw materials, like Qatar is for oil, Iceland for metals and fish, etc. Without international trade, these countries would never benefit from their natural resources or raw materials.

Greater choice for consumers: More international trade results in more choices of products.

Specialisation and economies of scale – greater efficiency: This means that it does not matter what a country is specialised in, and the essential thing is to pursue a specialisation that allows companies to make a profit that outweighs most of the other factors.

Global growth and economic development: International trade influences the economic growth of a country. This increase also leads to the reduction of poverty levels.

Income: It helps in earning foreign exchange to the organisations. Forex helps in paying off the cost of imports of capital goods, technologies, fertilisers, etc., from abroad.

Efficient resources: Under international trade, countries produce what they can produce efficiently and leave the other activities to nations in which they can work efficiently. This helps different nations to distribute the activities and work efficiently in their areas.

Growth and employment potentials: International trade helps in faster growth of organisations as well as countries. Sometimes, organisations are not able to create employment in the market as they produce on a small scale. Initially, countries like China, Japan, and South Korea took the whole world as a single market for trade. This helped them in employment generation across the world.

Standard of living: People in one country are able to enjoy goods and services of other nations. This helps them in improving the standard of living.

b) What are tariff barriers? Explain its impact on the economy.

A barrier to trade is a government-imposed restraint on the flow of international goods or services. Those restraints are sometimes obvious, but are most often subtle and non-obvious.

The most direct barrier to trade is an **embargo**– a blockade or political agreement that limits a foreign country's ability to export or import. Embargoes still exist, but they are difficult to enforce and are not common except in situations of war.

The most common barrier to trade is a **tariff**–a tax on imports. Tariffs raise the price of imported goods relative to domestic goods (good produced at home).

Another common barrier to trade is a government **subsidy** to a particular domestic industry. Subsidies make those goods cheaper to produce than in foreign markets. This results in a lower domestic price. Both tariffs and subsidies raise the price of foreign goods relative to domestic goods, which reduces imports.

Barriers to trade are often called “**protection**” because their stated purpose is to shield or advance particular industries or segments of an economy. From an economic perspective, though, the costs to the economy of reducing its opportunities to trade almost always outweigh the benefits enjoyed by those who are protected.

How Do Tariffs Impact the Economy?

Tariffs are imposed on an economy as taxes that either protect the domestic industry or heighten the economy's revenue. This is achieved by lowering the supply of foreign commodities and increasing their prices. The imposition of the tax leaves the burden of the tax on consumers to pay because it is reflected through an increment in prices. Therefore, when prices increase, the demand for the commodities falls, lowering their supply from foreign countries.

When prices rise, the consumers in the home country are the ones that feel the burden, and this translates to a decline in consumer surplus. However, the domestic producers are affected positively as they receive protection from cheap imports and higher prices due to the tariff's imposition. The domestic economy is protected as well as jobs due to an upswing in the market

share. However, the increased producer surplus is often lesser than the reduced consumer surplus, causing a welfare loss in the economy.

Generally, the domestic market is protected when domestic producers are hedged from foreign competition emanating from their cheap commodities. When cheap imports enter the country, consumers outrightly opt for cheap imports. Therefore, the government protects the domestic economy by imposing tariffs that heighten those prices, reducing their consumption.

20 a) What are the arguments in favour of free trade?

Free Trade

One view says that we should make it as easy as possible for goods and services to move between countries. This approach is based on the argument that more trade makes us wealthier and is therefore a good thing. It is known as *free trade*.

Another approach says that we should restrict trade. We might do this to protect certain jobs. We might think that we need certain industries – such as food production or steel-making – just in case things go wrong in the wider world. We might want to restrict imports from countries with lower labour or environmental standards so they can't undercut our industries. This approach is known as *protectionism*.

Many economists agree that some restrictions on trade are desirable, but that we should be careful, as such restrictions can make us poorer overall. For example, limits on agricultural imports may be good for British farmers, but they also increase food prices.

Arguments for Free Trade

There are several key arguments in favour of free trade:

- Free trade increases the size of the economy as a whole. It allows goods and services to be produced more efficiently. That's because it encourages goods or services to be produced where natural resources, infrastructure, or skills and expertise are best suited to them. It increases productivity, which can lead to higher wages in the long term. There is widespread agreement that rising global trade in recent decades has increased economic growth.
- Free trade is good for consumers. It reduces prices by eliminating tariffs and increasing competition. Greater competition is also likely to improve quality and choice. Some things, such as tropical fruit, would not be available in the UK without trade.
- Reducing non-tariff barriers can remove red tape, thus reducing the cost of trading. If companies that trade in several countries have to work with only one set of regulations, their costs of 'compliance' come down. In principle, this will make goods and services cheaper.
- In contrast, protectionism can result in destructive trade wars that increase costs and uncertainty as each side attempts to protect its own economy. Protectionist rules can tend to favour big business and vested interests, as they have the resources to lobby most effectively.

b) Explain absolute advantage theory with the help of an example

Absolute advantage is an economic principle that manifests when one company can create and distribute the same goods as another company, but with fewer assets. It refers to an organization's production level. One company's greater access to resources can make its design and manufacturing processes more efficient.

In international trade, countries specialize in producing a good to service the rest of the world. Suppose two countries want to exchange resources, and they both have absolute advantages over the other in the goods they're offering. The trade between them would be helpful to both countries

because they gained a resource they couldn't previously access. Examples of factors that fuel absolute advantage include:

- **Geography:** A country's location in the world can grant proximity to natural resources, which can give it an absolute advantage over other countries. For example, nations in the Middle East can produce oil more quickly and cost-effectively than countries in North America, empowering them with absolute advantage.
- **Technology:** Advanced technology can enable professionals to produce goods more quickly than competitors. Companies might have machines that can process and package products at more efficient rates than employees can by hand.
- **Wealth:** With high-income levels, a country or company can devote financial assets toward producing goods or offering services. They can also develop technology and hire more professionals, which grants them an absolute advantage over entities with less wealth.
- **Education:** Education can teach employees how to practice technical skills, which they can use to design products. A skilled workforce can streamline production for a company, offering an absolute advantage over a company whose employees have different expertise.

Example

A nation in the eastern part of the world has a robust mining industry. Its geographic location allows professionals to retrieve coal from beneath the earth's surface. The country's suitable environment for mining provides an absolute advantage for access to coal over other countries. It trades coal during international exchanges to gain other resources it needs, such as corn. In a northwestern country, the farmland is rich, so the growth of corn prospers in the region, providing an absolute advantage. Yet, there's a small demand for mining, and the supply of coal is low. Both countries enter an international exchange, where the eastern country gains corn and the northwestern country gains coal. The trade equally benefits the nations.